

Element H

We will test our design according to each of our four criteria as follows:

1. Load Time

1. Set up a timelapse or start stopwatch to use as a timer
2. Place a green tarp underneath the animal
3. Attach the engine lever to the fish hook on the ceiling
4. Carry the animal into the operating room underneath the project; move the animal/tarp as close as possible to be underneath the lift
5. Attach yellow straps on both sides to tarp handles using carabiners
6. Secure that all carabiners are safely attached and connected
7. Turn on the hook mechanism

If the animal can be placed on the lift with the lift turned on within our designated time frame (5-10 minutes), then it is successful.

2. Safety of the animal

1. Set up the project prior to loading the animal
2. Start the timer
3. Move the tarp underneath the animal, ensuring that it is fully supported and in an adequate position to be fully supported
4. With one person on each handle, move the animal on the tarp underneath the project
5. Attach the yellow straps to each handle of the tarp, making sure not to clip the animal itself
6. Slowly raise the fish hook to bring the tarp off the ground

7. Slowly lower the fish hook to bring the tarp to the ground

For this criteria to be met, the project should be able to load the animal and be able to balance out no matter where the heaviest spot of the tarp is.

3. Weight-bearing

To test for the amount of weight the project can lift and sustain for an extended amount of time we will:

1. Set up the tarp on the floor below the lift
2. Attach yellow straps to both sides of the tarp
3. One person lays in the center of the tarp along the length, make note of the person's weight
4. Lift the tarp up
5. Start timer; hold tarp for 15 minutes;

If the lift exceeds the 15-minute mark, bring the tarp back to the ground and add another person.

Continue to add weight til weight bearing reaches 1100 lbs (maximum weight). If a tarp can hold a minimum of 50 lbs and a maximum of 1100 lbs, it is successful.

4. Durability

1. Set up tarp and lift ready to use
2. On the tarp, lay weight straight along the length in the center of the tarp
3. Raise the tarp for 15 minutes
4. Push the tarp back and forth while in the air
5. Lower the tarp to the ground

6. Change weight to lay straight along the width
7. Raise the tarp for 15 minutes
8. Push the tarp back and forth while in the air
9. Lower the tarp to the ground

We will change the position of the weight, and the amount of weight per use, to make sure that wherever the weight is distributed, the center of mass does not alter and make one side of the tarp lean or waver. In this, we will make sure that the animal will be securely lifted into the air, with no possibility of losing balance, and if we achieve this, the weight bearing is successful.